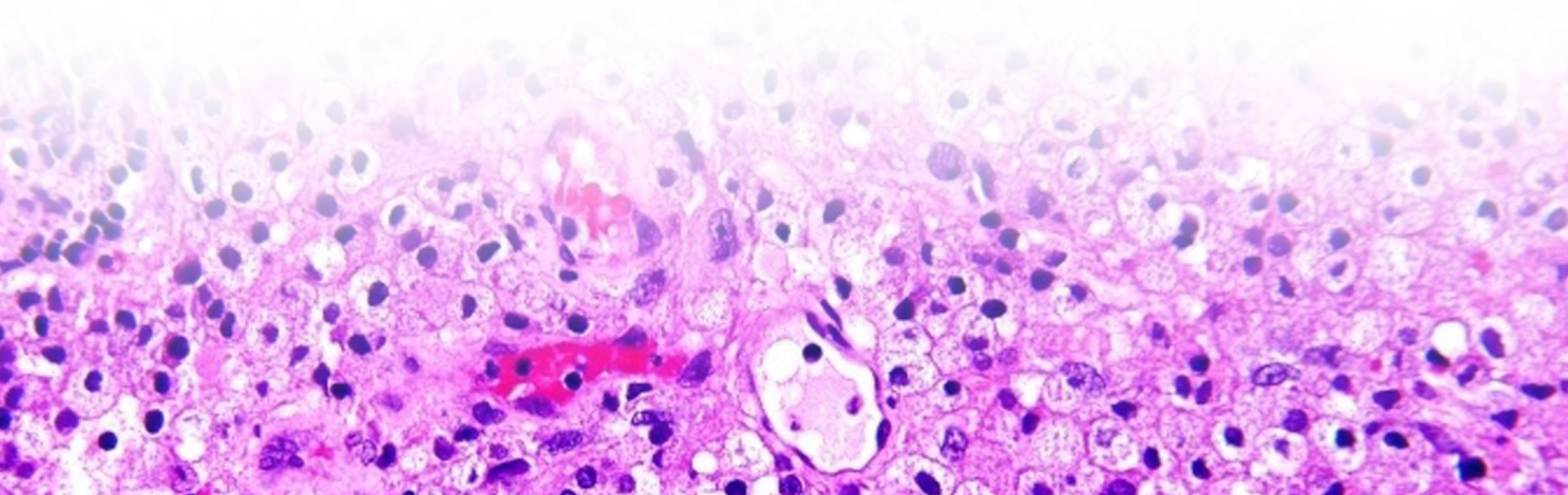
Student Notes

PATHOLOGY

1st Edition

Unit 4

Genetics



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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GENETIC DISORDERS

SHORT NOTES

1. Classification of Single gene disorders
2. Structural Abnormalities in chromosome
3. Autosomal Dominant Disorders
4. Autosomal Recessive Disorders
5. Sex-Linked disorders
6. G6PD
7. Familial hypercholesterolemia
8. Nieman-Pick disease
9. Gaucher's disease and the special stain to diagnose
10. FISH
11. Turner syndrome
12. Klinefelter syndrome
13. Trisomy 21
14. Mitochondrial inheritance

SHORT ANSWERS

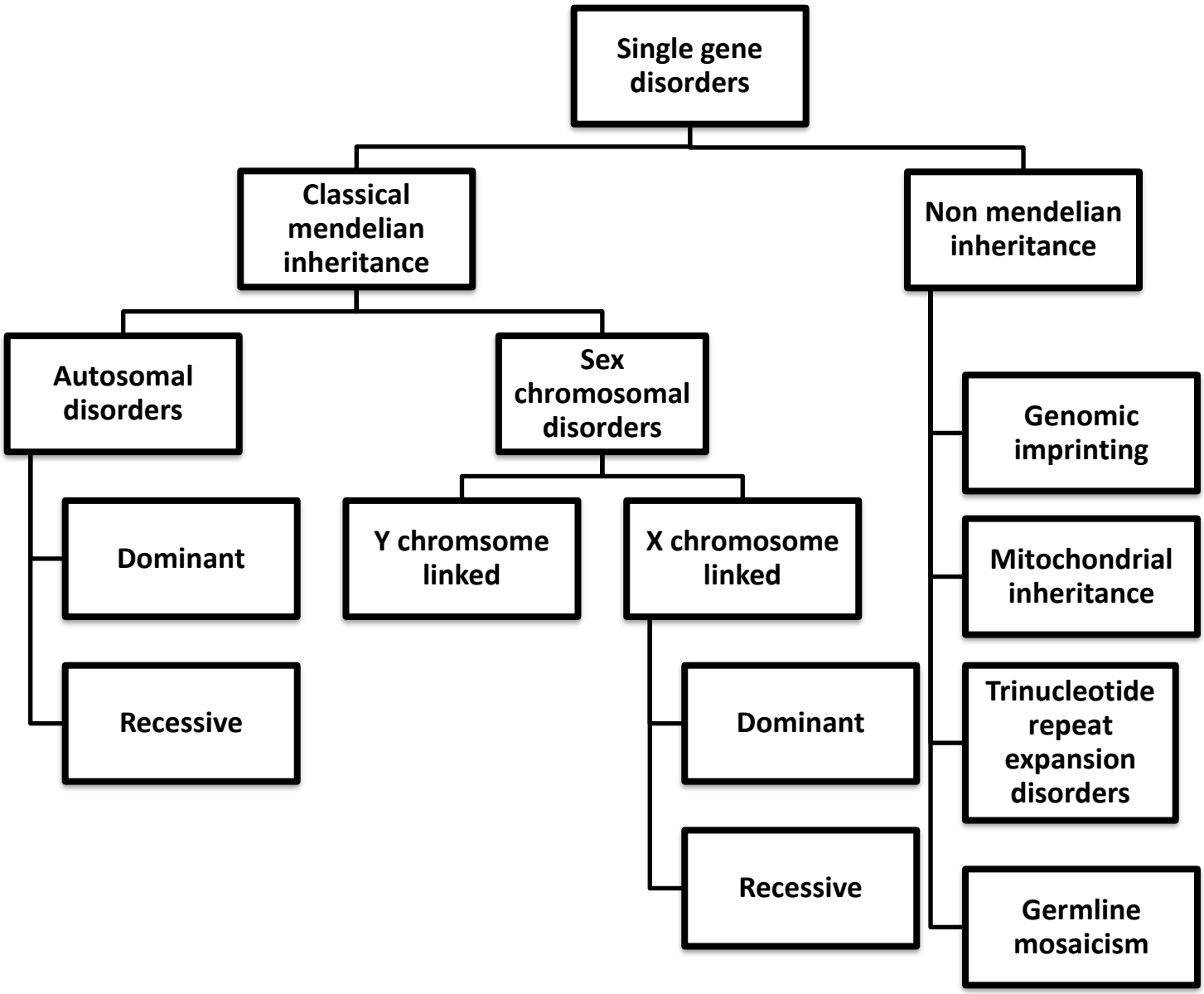
1. Autosomal Recessive hematopoietic diseases
2. Trisomy syndromes
3. Trisomy 18
4. Sex Chromatin

UPDATES

SHORT NOTES

1. CLASSIFICATION OF SINGLE GENE DISORDERS

- It is a **monogenic disorder.**



2. STRUCTURAL ABNORMALITIES ON CHROMOSOMES

BASED ON THE NUMBER OF CHROMOSOMES

- **Euploid** - multiples of 23.

- **Aneuploid**:
  - Error occurs in meiosis or mitosis - not an exact multiple of 23.
  - Due to non-disjunction and anaphase lag.
    - In Nondisjunction:
      → Either n+1 or n-1
      → Their fertilisation produces 2n+1 (trisomic) or 2n-1 (monosomic)
    - In Anaphase lag:
      → One homologous chromosome in meiosis or one chromatid in mitosis lags behind and left out of the nucleus.
  - Results in : **1 NORMAL CELL + 1 CELL WITH MONOSOMY**

- **Mosaicism:**
  - Mitotic errors (during the cleavage of the fertilised ovum or in somatic cells) in early development give rise to two or more populations of cells with different chromosomal complement in the same individual.
  - In division - 1 daughter cell → either 3 sex chromosome or only 1 sex chromosome.
  - For ex: either 47, XXX / 45, X - mosaic variant of Turner syndrome.

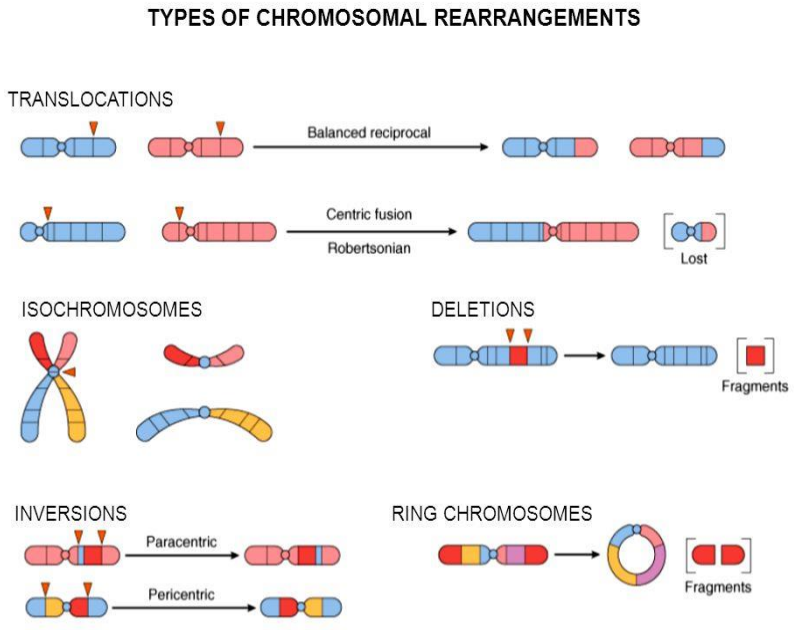
- **Autosomal mosaicism:**
  - Non-viable mosaic: error in early mitosis division affecting the autosomes.
  - Rarely this nonviable one is lost during embryogenesis which results in viable mosaic.
  - (ex:46, XY /47, XY, +21) - trisomy 21 mosaic with variable expressions of down syndrome

BASED ON THE STRUCTURE OF CHROMOSOME:

- **FISH (Fluorescence InSitu Hybridization)** can detect changes as small as kilo bases
- **Deletion:**
  - Loss of portion of a chromosome.
  - Both interstitial and rarely terminal.

  Types of Deletion:
  - **Interstitial:**
    - Occurs when there are 2 breaks within a chromosome arm, followed by loss of the chromosomal material between the breaks and fusion of the broken ends.
    - Eg: 46, XY, del (16) (p11.2p13.1) - breakpoints in the short arm (p) of chromosome 16 at 16p11.2 and 16p13.1.
  - **Terminal:**
    - Results from a single break in a chromosome arm, producing a fragment with no centromere, which is then lost at the next cell division, and a chromosome bearing a deletion.
    - **Note**: End of the chromosome is protected by acquiring telomeres sequences.

- **Ring chromosome:**
  - Break occur at both ends of chromosome with fusion of damaged ends.
  - Results in serious consequences.
  - Example: 46, XY, r(14)



- **Inversions:**
  - Two breaks within a single chromosome with reincorporation of the inverted, intervening segment.
  - Involving one arm - paracentric.
  - Breaks are on the opposite sides of the centromere - pericentric.

- **Isochromosome:**
  - When one arm of a chromosome is lost and the remaining arm is duplicated, results in a chromosome consisting of 2 shots arms only or of 2 long arms.
  - **Note**: Isochromosome has morphologically identical information in both arms.
  - Most common - live births
  - Eg: i(X)(q10) involves long arm of the X.
    - Xq isochromosome associated **Monosomy** - for genes on the **short arm of X**.
    - Xq isochromosome associated **Trisomy** - for genes on the **long arm of X**.

- **Translocation:**
  - Segment of 1 chromosome is transferred to another.
  - Types:
    - Balanced reciprocal
    - Robertsonian (centric fusion)

  - **Balanced reciprocal:**
    - Single breaks in each of two chromosomes with exchange of material.
    - Ex: 46, XX,t(2;5)(q31;p14) – Translocation between the q of chromosome 2 and p of chromosome 5: subsequent fertilization of the normal gamete would lead to spontaneous abortion or birth of a malformed child.

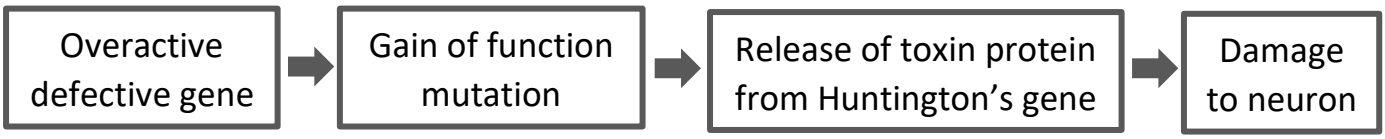
  - **Robertsonian:**
    - A translocation between two acrocentric chromosomes.
    - 1 in 1000 occurrence.
    - Leads to 1 very large chromosome and 1 extremely small chromosome.
    - Significance lies in the production of abnormal progeny as in down's syndrome.

3. AUTOSOMAL DOMINANT DISORDERS:

- They can express themselves only in a heterozygous state.
- Affects **50**% of the progeny.
- One parent is affected in each generation.
- And such inheritance is called as **vertical inheritance.**
- Affecting structural proteins mostly.
- Remember any genetic mutation has two possibilities:
    - **Underactivity** of the genes leading to loss of function mutation.
    - **Overactivity** of genes leading to gain of function mutation.
- Autosomal dominant disorder – majority has loss of function mutation and affecting structural proteins.

EXAMPLES:

- Von Willebrand's disease/von Hippel Lindau syndrome.
- Familial hypercholesterolemia/familial adenomatous polyposis
- Polycystic kidney disease.
- Osteogenesis imperfecta
- Marfan's syndrome
- Intermittent porphyria.
- Neurofibromatosis.
- Achondroplasia.
- Multiple endocrine neoplasia.
- Tuberous sclerosis.
- Huntington's disease.



- Hereditary spherocytosis.

PROPERTIES:

- **Incomplete penetrance:**
  - Every individual having the defective gene will not show any clinical manifestations.
  - Reasons are –
    - Depends on the environmental factors.
    - Except the defective gene, the normal gene can produce some proteins which is sufficient for the tissues/cells (as it usually affects structural proteins).
- **Variable expressivity:**
  - The affected individuals express their clinical manifestations as – mild / moderate / severe.
  - That is the symptoms are variable.
- **Pleiotropy:**
  - Ex. Neurofibromatosis.

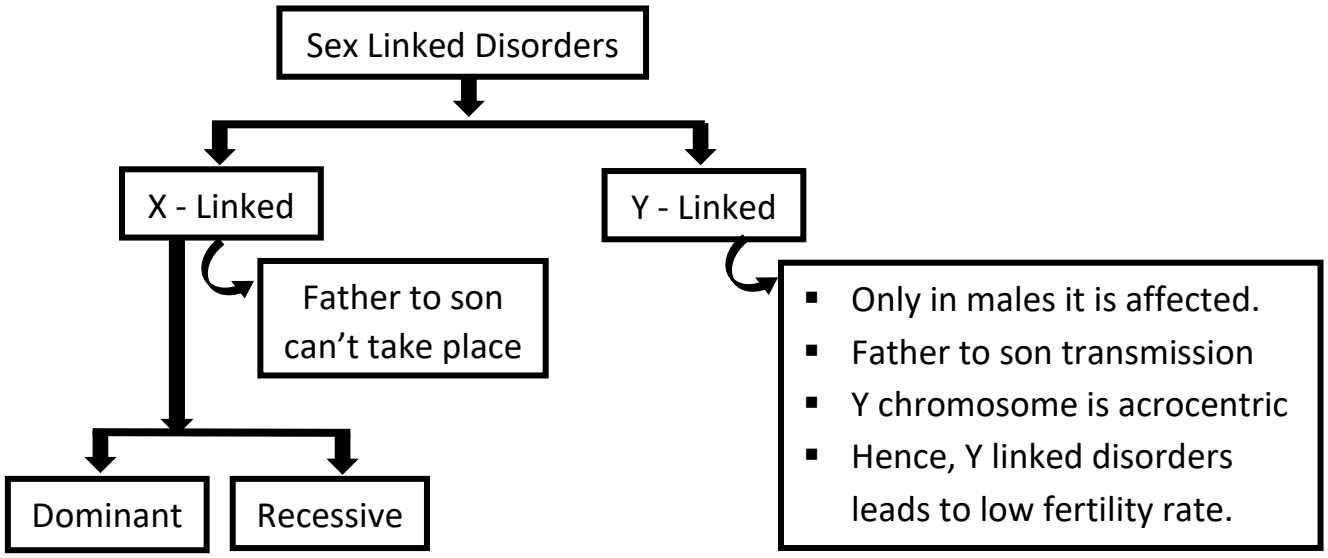
4. AUTOSOMAL RECESSIVE DISORDERS:

- Involvement of enzymatic proteins mostly seen in cases of inborn errors of metabolism.
- Expressed only in the homozygous state.
- Have an early onset of clinical manifestations due to homozygous condition.
- **Complete penetrance** - all persons who are affected shows the clinical manifestations.
- Common in consanguineous marriage.
- Here parents of the patient act as carriers and the siblings are affected and this type of inheritance is called as **horizontal inheritance.**

EXAMPLES:

- Friedreich's ataxia.
- Sickle cell anemia.
- Thalassemias.
- Wilson disease in which the copper metabolism is affected.
- Hemochromatosis in which the iron metabolism is affected.
- Homocystinuria.
- Alkaptonuria.
- Inborn errors are
  - Glycogen disorder.
  - Lysosomal disorders.

5. SEX LINKED DISORDERS:



X LINKED DOMINANT DISORDER:

- Most common sex-linked inheritance.
- X linked gene coding for enzymatic genes.
- Males are more commonly affected.
- Females act as carriers as they are heterozygous
- Females are going to be affected only under two conditions:
  - Homozygous condition.
  - When one of the normal X chromosomes in-activated.
- Examples
  - Alport syndrome.
  - Vitamin D resistant rickets.
  - Incontinentia pigmenti.

X-LINKED RECESSIVE DISORDER:

- Affecting male individuals.
- Transmitting to daughters and they act as carriers (x'x)
- Involvement of structural proteins is less common.
- Examples:
  - Lesch Nyhan syndrome.
  - Haemophilia A/B.
  - Colour blindness and the commonest colour affected is red.
  - Chronic granulomatous condition.
  - G6PD deficiency.
  - Duchenne muscular dystrophy.
  - Agammaglobulinemia (Bruton's disease)
  - Fragile X syndrome.
  - Wiskott Aldrich syndrome.

6. GLUCOSE-6-PHOSPHATE DEHYDROGENASE (G6PD) DEFICIENCY

- Hereditary- X linked Recessive
- Increased incidence in males.

PATHOGENESIS:

Deficient / impaired enzyme function
↓
Abnormalities in HMP shunt / glutathione metabolism
↓
Decreased protection of RBCs against oxidative injuries
↓
Hemolysis

- **Characteristic feature:** Episodic hemolysis caused by exposure to oxidant stress
- **Results in**:
  - Intravascular and Extravascular hemolysis.
  - **Heinz bodies**: Dark inclusions within RBCs stained with crystal violet
  - Heinz bodies → membrane damage → intravascular hemolysis

COMMON TRIGGERS:

- Infections (Viral Hepatitis, Pneumonia, Typhoid Fever)
- Drugs(Anti-malarials, Sulfonamides, nitrofurantoins)
- Food(Fava bean)

FATE OF CELLS:

- **Bite cells:** RBCs with Heinz bodies → pass though splenic cords → macrophages pluck out Heinz bodies → abnormal shape like a bite taken out of the cell.
- **Spherocytes:** Due to loss of membrane surface area → Cells are destroyed by phagocytes

SYMPTOMS:

- G6PD deficient individuals on exposure to oxidants(after 2-3 days)
- Anaemia,
- Hemoglobinemia,
- Haemoglobinuria.
- Features related to chronic hemolysis-absent (because of intermittent hemolysis)

TREATMENT:

- Self-limited as only older RBCs are at risk for lysis.

7. FAMILIAL HYPERCHOLESTEROLEMIA

- Mendelian disorder
- Receptor defect due to mutation in gene encoding the receptor for LDL
- Receptor defect → loss of feedback control → premature atherosclerosis → ↑risk of MI

NORMAL PROCESS OF CHOLESTEROL METABOLISM AND TRANSPORT:

i. Liver secretes VLDL into blood.
ii. VLDL is cleaved into IDL in the capillaries by lipoprotein lipase.
iii. 50% of IDL is taken into liver by receptor mediated transport through LDL receptor
iv. And the left over IDL is converted into LDL after removing most of triglycerides.
v. LDL removal:
    a. Mediated by LDL receptor.
    b. Receptor for oxidized LDL.
vi. After a series of process, the LDL molecule is enzymatically degraded by lysosomes into amino acids and free cholesterol.
vii. Functions of intracellular cholesterol: Suppresses further cholesterol synthesis, Favors storage of excess cholesterol and suppresses synthesis of LDL receptors.

PATHOGENESIS:

Since this is a LDL receptor defect there is:

- Defective LDL clearance.
- Increased LDL levels in blood(since IDL uses LDL receptor,its impairment diverts a greater proportion of IDL into forming plasma LDL pool)

In hypercholesterolemia, there is increase in the scavenger mediated traffic of LDL cholesterol into the cells of mononuclear phagocyte system and the vascular walls (causing xanthomas & premature atherosclerosis)

MOLECULAR GENETICS INVOLVING THE LDL RECEPTOR GENE:

- **Class I mutations**: Complete failure of synthesis of receptor protein.
- **Class II mutations:** Receptor proteins accumulate in ER because of folding defects.
- **Class III mutations**: Defect in the LDL binding domain of the receptor.
- **Class IV mutations**: Failure to localize in coated pits,hence LDL is not internalized.
- **Class V mutations**: Defect in dissociation of receptor and bound LDL inside the cell.

TREATMENT: Statins-suppress intracellular cholesterol synthesis.

8. NIEMANN-PICK DISEASE

TYPES- A and B

- Autosomal recessive.
- Inherited deficiency of sphingomyelinase → Lysosomal accumulation of sphingomyelin.

Type A (Severe Infantile form)	**Type B**
Neurologic involvement	Organomegaly with no CNS involvement
Death within first 3 years of life	Survives into adulthood

MORPHOLOGY:

- Affected cells enlarges (upto 90 micrometer in diameter) due to distension of lysosomes with sphingomyelin and cholesterol.
- Brain- Gyri shrunken and sulci widened.
    - Vacuolation and ballooning of neurons.
    - **Retinal cherry red spot**.
- Microscopically: Small vacuoles of uniform size.
    - Foam cytoplasm.
    - Vacuoles stain for fat.
    - **Zebra bodies**.

CLINICAL FEATURES:

- Protuberant abdomen (due to splenomegaly and hepatomegaly)
- Failure to thrive,
- Vomiting,
- Fever,
- Generalized lymphadenopathy.
- Progressive deterioration of psychomotor function.

TYPE-C

- Primary defect in non-enzymatic lipid transport.
- Mutation in genes- NPC1(95% of cases) & NPC2.
- Functions of NPC genes: transport of free cholesterol from lysosomes to cytoplasm.
- Presents as:
    - Hydrops fetalis & stillbirth,
    - Neonatal hepatitis,
    - Progressive neurologic damage.

9. GAUCHER'S DISEASE

- Lysosomal storage disorder.
- Cluster of autosomal recessive disorders due to mutation in gene encoding glucocerebrosidase (enzyme that cleaves glucose residue from ceramide)
- **Results in:** Accumulation of glucocerebroside in phagocytes & CNS.
- **Source of glucocerebroside**:
  Senescent leukocytes, RBCs → glycolipids catabolism → glucocerebroside
- **Stain to diagnose gaucher's disease**-Periodic acid Schiff (PAS) stain-positive.

CLINICAL SUBTYPES:

- **Type 1**: Most common (99%) (systemic involvement)
  - Chronic non-neuronopathic form.
  - Storage only in mononuclear phagocytes, no CNS involvement.
  - Splenic and skeletal involvement.
  - Decreased glucocerebrosidase activity.
- **Type-2:** Acute non-neuropathic form (infantile acute cerebral pattern)
  - Hepatosplenomegaly & progressive CNS involvement.
  - Absence of glucocerebrosidase activity.
- **Type-3:** Intermediate btw type-1 and type-2

MORPHOLOGY

- **Gaucher cells** - distended phagocytes found in Lymphoid organs
- Fibrillary type of cytoplasm.
- Enlarged spleen (up to 10kg)
- Lymphadenopathy: mild to moderate.
- Accumulation of Gaucher cells in BM → area of bone erosion → ↑ chances of fracture.
- In CNS involvement, Gaucher cells are seen in Virchow robin spaces
  - Neurons appear shriveled & progressively destroyed.

CLINICAL FEATURES:

- I: Splenomegaly, Bone Involvement, Pancytopenia/Thrombocytopenia.
- II, III: CNS dysfunction, convulsions, progressive mental deterioration.

TREATMENT:

- Replacement therapy.
- Allogenic hematopoietic stem cell transplantation.

10. FISH:

TECHNIQUES FOR MOLECULAR ANALYSIS OF GENOMIC ALTERATIONS

- Fluorescence in situ hybridization (FISH).
- Multiplex ligation-dependent probe amplification (MLPA).
- Southern blotting.
- Cytogenomic Array Technology.
- Array-Based Comparative Genomic Hybridization (Array CGH).

FLUORESCENCE IN SITU HYBRIDIZATION (FISH).

- Uses DNA probes that recognize sequences specific to particular chromosomal regions.
- Extension of FISH is chromosome painting.
- Fluorescent dyes - emit different wavelengths.
- FISH can be performed on:
    - Prenatal samples.
    - Peripheral blood cells.
    - Touch preparations from cancer biopsies.
- Used to detect:
    - Aneuploidy,
    - Subtle microdeletion,
    - Complex translocation,
    - Gene amplification - Eg:
        - HER2 in breast cancer
        - NMYC amplification in neuroblastomas.

KARYOTYPING - SPECTRAL (MULTICOLOUR FISH)

- 5 fluorochromes – Permits visualization of entire human genome
- Well called as “Spectacular karyotyping”.

11. TURNERS SYNDROME

- HYPOGONADISM IN FEMALES
- Mainly 45, X karyotype
- Also present with partial monosomy
    - Loss of short arm,
    - Loss of both short and long arm,
    - Deletion of a portion of short or long arm.
- Sometime mosaic patients are also seen.

FEATURES:

- Edema of dorsum of hand and foot (due to lymph stasis)
- Swelling of nape of neck
- **Neck webbing**
- Congenital heart disease
- Coarctation of aorta
- Bicuspid aortic valve
- Inadequate breast development
- Decreased secondary sexual characters
- Little pubic hair
- Defect in visual –spatial information
- Shortness of stature
- Amenorrhea
- Hypothyroidism
- Rarely glucose intolerance, obesity
- **Streak ovaries** (accelerated loss of oocytes by the age of 2)

12. KLINEFELTER SYNDROME

- Male hypogonadism .
- Mainly pattern of 47,XXY
- Occur due to nondisjunction of chromosomes in meiosis of paternal or maternal side.
- Maternal age increases the risk.
- Mosaic patterns can also be seen.
- Symptoms appear after puberty.
- Occur also due to escape of X linked genes from X INACTIVATION and over express.
- Longer CAG receptors on androgen coding gene also causes less sensitivity to androgen.

FEATURES

- Increased length in between sole and pubic bone.
- Small atrophic testes.
- Small penis.
- Lack of secondary sexual characters like deep voice, beard and pubic hair.
- Gynecomastia.
- Lower IQ but normal.
- Type 2 diabetes.
- Mitral valve prolapse.
- Osteoporosis.
- Reduces spermatogenesis.
- Infertility.
- Breast cancer.
- Systemic lupus erythematosis.
- Extra gonadal germ cell tumor.
    - FSH level increased and TESTOSTERONE reduced.

13. TRISOMY 21

- Mental retardation is the prominent (severe)
- It is called as DOWN SYNDROME
- Maternal age has a strong influence.

PATHOGENESIS

- Non-dysfunction in the first meiotic division of gametogenesis (95%)
- Robertsonian translocation (4%)
- Mosaics (1%)

CLINICAL FEATURES

- Mentally retarded
- Flat facial profile.
- Oblique palpebral fissures and epicanthic folds.
- Enlarged and malformed ears
- Protruded tongue
- Abundant neck skin
- **Simian crease**
- 40%-congenital heart disease.
- Oesophageal atresia.
- Intestinal stenosis
- Some may develop leukemia.
- Abnormal immune response infections.
- Neurological changes in patients older than age 40 develop Alzheimer diseases.

DIAGNOSIS

- By cranio-facial appearance at time of birth
- Can be done with great precision using Next Generation Sequencing

14. MITOCHONDRIAL INHERITANCE

- The human mitochondrial DNA contain 37 genes of which 22 are transcribed to tRNA and 2 into r-RNA.
- The remaining 13 genes encodes for respiratory chain enzymes. (Since mtDNA participates in oxidative phosphorylation.)

CHARACTERISTICS:

- The mitochondrial DNA has a unique feature of maternal inheritance. (this is because ova contains abundant mitochondria in the cytoplasm)
- The mutations in mtDNA show its effect in CNS, skeletal muscle, cardiac muscle, liver and kidneys.

CLINICAL SIGNIFICANCE:

- The individual who carries both normal mtDNA and mutant mtDNA is called **Heteroplasmy.**
- During cell division the mitochondrial DNA are distributed to daughter cells resulting in variable mtDNA in daughter cells.
- The mutant mitochondrial gene mainly affects the neuromuscular system i.e. **Leber hereditary optic neuropathy.**

SHORT ANSWERS

1. AUTOSOMAL RECESSIVE HAEMATOPOEITIC DISORDERS

- Sickle cell anemia.
- Thalassemias

2. TRISOMY SYNDROMES

- Trisomy 21 (Down syndrome)
- Trisomy 18 (Edwards syndrome)
- Trisomy 13 (Patau syndrome)
- TRISOMY involving chromosomes 8,9,22

3. TRISOMY 18

The trisomy 18 is called Edwards syndrome which occurs due to meiotic non-disjunction resulting in extra copy of chromosome.

FEATURES:

- Prominent occiput.
- Mental retardation.
- Micrognathia.
- Low set ears.
- Overlapping fingers.
- Congenital heart defects.
- Renal malformations.
- Rocker bottom feet.

4. SEX CHROMATIN

- Inactivated X CHROMOSOME in cell with more than ONE X CHROMOSOME.
- Usually present in females.
- By a process called **lyonization**.
- They are also called Barr bodies.
- They appear as dark staining spots on the periphery of the nucleus of somatic cells

UPDATES FROM ROBBINS: 10TH EDITION

1. FRAGILE X-ASSOCIATED PRIMARY VARIAN FAILURE

- CGG prematuration in FMR 1 gene is seen in Fragile X-associated primary ovarian failure, as well as in Fragile X-ataxia (progressive neurodegenerative syndrome)
- 20% of women (carriers) having the prematuration develop ovarian failure (below the age of 40)
- Women affected with fragile X-associated primary ovarian failure have menstrual abnormalities and decreased fertility
- FSH levels are elevated, and anti-mullerian hormone levels are decreased – both are markers of declining ovarian function.
- They develop menopause approximately 5 years earlier than the controls.
- Aggregates containing FMR1 mRNA are detected in granulosa and ovarian stromal cells.
- The aggregates are responsible for the premature death of ovarian follicles

3. TGF-β BIOAVAILABILITY IN MARFAN SYNDROME:

- Fibrillin-1 controls the bioavailability of TGF-β.
- Reduced or altered forms of fibrillin-1 give rise to abnormal and excessive activation of TGF-β, since normal micro fibrils sequester TGF-β.
- Excessive TGF-β signalling leads to:
  - Inflammation,
  - Has deleterious effects on vascular smooth muscle development
  - Increases the activity of metalloproteases → loss of extracellular matrix.
- The finding is supported by following observations:
  - Gain-of-function mutations in the **TGF-β type II receptor** leads to **Marfan syndrome type 2 (MFS2)**.
  - Patients with germline mutations in one isoform of TGF-β, called **TGF-β3**, present with an **inherited predisposition to aortic aneurysm** and other cardiovascular manifestations similar to those found in patients with classic Marfan syndrome.
  - **Angiotensin receptor II blockers**, which inhibit TGF-β activity, reduce the aortic root diameter in mouse models of Marfan syndrome.

4. MITOPHAGY

- Turnover of mitochondria by autophagy is called mitophagy
- This serves as a quality control system where dysfunctional mitochondria are degraded

5. ADDITIONAL MUTATIONS IN FAMILIAL HYPERCHOLESTEROLEMIA

- Mutations in **gene encoding ApoB**.
  - ApoB on the surface of LDL particles is the ligand for LDL receptors
  - Thus mutant ApoB reduces the binding of LDL molecules with LDL receptors.
  - This compromise increases serum LDL cholesterol.
- Activating mutation in the **PCSK9 gene**.
  - It greatly reduces the number of LDL receptors on the cell surface because of their increased degradation during the recycling process.

6. TAY-SACHS DISEASE:

- There are two isoenzymes of β-hexosaminidase:
  - Hex A, consisting of two subunits, α and β
  - Hex B, a homodimer of β-subunits.
- Degradation of GM2 gangliosides requires 3 proteins encoded by three distinct genes:
  - HEXA (on chromosome 15) encodes the α-subunit of Hex A;
  - HEXB (on chromosome 5) encodes the β-subunit of Hex A and Hex B;
  - GM2A (on chromosome 5) encodes the activator of hexosaminidase

7. GAUCHERS AND PARKINSONS:

- Mutation of the glucocerebrosidase gene is the most common known genetic risk factor for development of Parkinson disease.
- Patients with Gaucher disease have a 20-fold higher risk of developing Parkinson disease (compared with controls), and 5% to 10% of patients with Parkinson disease have mutations in the gene encoding glucocerebrosidase.
- There is a reciprocal relationship between the level of glucocerebrosidase and aggregation of α-synuclein, the protein involved in of Parkinson disease.
- Patients with lysosomal diseases have impairment in autophagy, leading to secondary accumulation of autophagic substrates including ubiquitinated and aggregate-prone polypeptides such as α-synuclein and Huntingtin protein.

8. VARIANTS OF MUCOPOLYSACCHARIDOSES:

- **09th edition:** 7 clinical variants
- **10th edition:** 11 clinical variants

9. CONGENITAL HEART DISEASE IN DOWN SYNDROME:

- Atrioventricular septal defects (43%) – the most common defect
- Ventricular septal defects (32%)
- Atrial septal defects (19%)
- Tetralogy of Fallot (6%)

10. RISK OF LEUKEMIA IN DOWN SYNDROME

- 20-fold increased risk of acute B lymphoblastic leukemias
- 500-fold increased risk of acute myeloid leukemias.

11. GENE SAMPLING OBSERVATION IN DOWN SYNDROME:

- **Gene dosage**.
  - Most of the protein coding genes mapped to chromosome 21 are overexpressed.
  - Eg: **Gene for amyloid-beta precursor protein (APP)**
    - ✓ Aggregation of amyloid-beta proteins is the critical initiating event in the development of Alzheimer disease
    - ✓ This contributes to the early onset of Alzheimer disease in Down syndrome.
- **Mitochondrial dysfunction.**
  - Approximately 10% of the genes overexpressed in Down syndrome are involved in regulation of mitochondrial functions.
  - Thus mitochondria are abnormal both morphologically and functionally in several tissues. For example:
    - ✓ Cristae are broken or swollen
    - ✓ There is evidence of mitochondrial stress with generation of ROS and activation of apoptosis.
- **Noncoding RNAs**.
  - Chromosome 21 has the **highest density of lncRNAs** whose target sequences are widely expressed on other chromosomes
  - Thus the process of identification of the genes become cumbersome

12. FACTORS INVOLVED IN KLINEFELTER SYNDROME:

- The clinical features of Klinefelter syndrome can be attributed to two major factors:
  - Aneuploidy and the impact of increased gene dosage by the supernumerary X
  - Presence of hypogonadism.

13. ROLE OF SHOX GENE IN LINEFELTER SYNDROME:

- The **Short-stature HomeobOX (SHOX) gene** maps on the pseudo-autosomal region of Xp
- It is one of the genes that is not subjected to X inactivation.
- An extra copy of this growth-related gene is responsible for the tall stature and long legs typical of Klinefelter syndrome.

14. TURNER SYNDROME AND CARDIOVASCULAR ABNORMALITIES:

- Approximately 5% of young women initially diagnosed with coarctation of aorta have Turner syndrome.
- Aortic root dilatation is present in 30% of cases
- There is a 100-fold higher risk of aortic dissection.

15. ROLE OF SHORT AND LONG ARMS IN TURNER SYNDROME:

- Based on the studies of patients with Turner syndrome with deletions affecting the short or long arm , it is shown that,
  - Somatic features are determined by genes on the short arm
  - Genes on the long arm affect fertility and menstruation.

16. INTELLECTUAL DISABILITY:

- Fragile X Syndrome – the most common genetic cause
- Down Syndrome - the most common cause

17. TNBX GENE:

- TNBX gene encodes for tenanscin-X, a large multimeric protein
- It interacts with fibrillar types I, III, and, V collagens
- Mutation of TNBX gene causes classic EDS-like condition.

18. BASIC STEPS OF BIOINFORMATICS

- **Alignment.**
  - Here sequencing reads from a sample are mapped onto a reference genome
  - This allowing them to be viewed and interpreted in context.
- **Variant calling**.
  - This process involves a systematic comparison of all of the sequence data from a sample with the reference sequence.
  - The more reads that cover a particular location (sequencing depth), the more likely that a variant will be detected if present.
  - If a locus shows sufficient evidence of a difference from the reference sequence, a variant call is made.
- **Variant annotation and interpretation.**
  - Called variants can be annotated with various features:
    - ✓ Gene name
    - ✓ Coding change and protein effect predictions
    - ✓ Information from databases listing benign and pathogenic variants
    - ✓ Clinical information
  - Once completed by the clinical laboratory, the annotated data are ready for reporting.
  - An important part of this report is a brief description of the biological and clinical significance of each pathogenic variant, which can be numerous in the case of some cancers.
- **Mutational signature calling.**
  - In addition to totalling up individual mutations, informatics algorithms have been developed that detect patterns of mutations that point to particular environmental exposures (e.g., ultraviolet light) or underlying defects in DNA repair.
  - This has become clinically relevant with the discovery that cancers with acquired defects in mismatch repair genes that lead to **microsatellite instability (MSI)** and are highly responsive to immune checkpoint inhibitors.
  - Based on this observation, clinical laboratories are now “calling” the MSI status of cancers based on NGS results.

19. HYBRID CAPTURE

- It is one of the sample preparation for targeted sequencing
- It is done by enriching for sequences of interest via custom complementary probes

20. ONE LINERS

- The most common extragonadal tumour in Klinefelter syndrome is **mediastinal teratoma**
- Gene encoding Lysyl oxidase is **PLOD1**, whose mutation results in Ehler-Danlos syndrome

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